ClapCap: CLAP Prefix for Music Captioning

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***Abstract*—A useful music caption names genre, instrumentation and mood rather than discrete sound events. We ask how far a lightweight language model can go when its only view of the audio is a short learned prefix projected from a frozen embedding. CLAPCAP is the audio analogue of the ClipCap image-captioning recipe: a frozen CLAP encoder produces one clip embedding, a small MLP maps it to a fixed-length prefix of pseudo-tokens, and a pretrained language model decodes the caption. On MusicCaps we compare GPT-2 (124M, decoder-only) with T5-base (220M, encoder–decoder), holding the encoder, projection, data split and evaluation fixed, and we measure a per-model noise floor over three seeds so that every ablation delta is judged against run-to-run variance. The two models are statistically indistinguishable at baseline (FENSE 0.400 vs. 0.395; noise floors ± 0.014 and ± 0.035), and second-stage fine-tuning erases the differences introduced in the projection. The largest decoding lever lifts FENSE to 0.580 almost entirely by removing the fluency-error term rather than by improving semantics. Swapping the general frontend for music-specialised encoders helps only where pretraining is pooled and joint audio–text. Against zero-shot reference captioners scored on the identical split, the pipeline is the strongest leakage-free system, and a pairwise human and LLM-as-judge study recovers a GPT-2-over-T5 preference that the automatic metrics call a tie. Every experiment runs on a single consumer GPU.**

***Index Terms*—Music captioning, audio–language models, prefix conditioning, CLAP, controlled ablation, evaluation metrics**

I. INTRODUCTION

Most audio captioning research targets *general* sound — environmental events and acoustic scenes — on AudioCaps [1] and Clotho [2]. Music is different: a useful music caption describes genre, instrumentation, mood, tempo and structure rather than discrete events. This work asks a deliberately minimal question: can a lightweight language model recover music semantics from a simple projection of a frozen audio embedding?

Our approach, CLAPCAP, adapts ClipCap [3], which captions images by projecting a frozen CLIP embedding into a prefix of pseudo-tokens for GPT-2. We replace the image encoder with a frozen CLAP audio encoder [4], caption music instead of images, and turn the single-model recipe into a *controlled comparison*: because the encoder, projection, data split and evaluation are shared, the language model is the only thing that changes between conditions. The contributions are:

- an audio adaptation of the prefix-captioning recipe to music, with a *noise-floor* methodology — per-model run-to-run variance from three seeds — so that single-run ablation deltas are interpretable rather than anecdotal;
- a controlled comparison of a decoder-only (GPT-2) and an encoder–decoder (T5) language model, with ablations over projection architecture, decoding strategy and the frozen audio frontend, each judged against that floor;
- a leakage-aware comparison against zero-shot reference captioners scored on the identical test split and harness, separating honest scores from training-on-test artifacts; and
- a pairwise human and LLM-as-judge study measuring agreement across lay listeners, audio-grounded models and text-only models.

We do not claim state of the art. Published captioning systems are evaluated on different datasets with different reference counts, so a numerical comparison across them would be meaningless; the contribution is the controlled analysis itself, together with the efficiency of the recipe — only a small projection, plus the language model, is trained.

II. RELATED WORK

Prefix captioning. ClipCap [3] maps a frozen CLIP image embedding to a constant-length prefix consumed by GPT-2, training only a small mapping network. It offers a lightweight transformer mapper with a frozen LM and an MLP mapper with a fine-tuned LM; CLAPCAP is the audio analogue of the latter. Notably, ClipCap reports that this variant *overfits* as trainable capacity grows — an observation we recover, amplified, when scaling the LM. In the audio domain, Pengi [5] likewise conditions a small LM on an audio-derived prefix, framing audio tasks as text generation.

Music captioning. MusicCaps [6] is a hand-curated set of 5,521 ten-second clips from AudioSet [7], each annotated by musicians. LP-MusicCaps [8] addresses caption scarcity by generating pseudo-captions with an LLM over tag datasets. Such domain captioners, together with BLAP [9], are a natural *reference* for CLAPCAP, and we score them on our exact split in Section IV-G.

Audio language models. End-to-end audio LLMs such as Qwen2-Audio [10], Audio Flamingo [11] and SALMONN [12] couple a large LM to an audio encoder and are trained on large instruction corpora. They are far heavier than the present recipe and bypass the frozen-encoder/learned-prefix design we study, so we treat them as zero-shot references rather than controlled comparisons.

III. Method

A clip is encoded once by a frozen CLAP model into a d_a-dimensional embedding. A projection network f_θ maps this vector to a prefix $p_1, \ldots, p_k \in \mathbb{R}^{d_{\text{lm}}}$ of k pseudo-tokens in the LM embedding space, and the LM generates the caption autoregressively, conditioned on that prefix (Figure 1).

Audio encoder. We use the LAION checkpoint `laion/clap-htsat-unfused` [4], whose audio tower is an HTS-AT transformer [13] producing a 512-dimensional embedding. CLAP is trained on general audio and is *not* music-specialised — a deliberate test of how far a general frontend carries on a music task, which Section IV-E then probes directly. Embeddings are precomputed once and cached, so the encoder never runs during training or evaluation.

Projection and language models. The projection is a small MLP, Linear → LayerNorm → GELU → Dropout → Linear, reshaped to (k, d_{lm}), with baseline prefix length k=8, dropout 0.3 and depth 2. We compare GPT-2 [14] (decoder-only, 124M) and T5-base [15] (encoder–decoder, 220M); for T5 the prefix is fed to the encoder, for GPT-2 it is prepended as input embeddings.

Two-stage training. In Stage 1 the LM is frozen and only the projection is trained; in Stage 2 the projection and the full LM are fine-tuned together, the audio encoder staying frozen. These correspond to ClipCap's frozen-LM and fine-tuned-LM regimes. The objective in both stages is autoregressive cross-entropy on caption tokens conditioned on the prefix,

$$\mathcal{L} = -\sum_j \log p_\theta(c_j \mid p_1, \ldots, p_k, c_1, \ldots, c_{j-1}) \, .$$

We use AdamW [16] (weight decay 0.1), cosine annealing [17], batch size 8 and early stopping (patience 5). Per model only the learning rates are tuned, via a 16-trial Optuna [18] search over an eight-epoch proxy; everything else is fixed so the LM stays the controlled variable.

Efficiency. The only newly added module is the projection (∼10.2M parameters). The complete model is 124M (GPT-2) or 220M (T5) parameters — one to two orders of magnitude smaller than billion-parameter end-to-end audio LLMs — and every experiment in this paper runs on a single consumer GPU (NVIDIA RTX 5090).

IV. Experimental Study

A. Setup

All experiments run on one workstation with an NVIDIA RTX 5090, using PyTorch, HuggingFace `transformers`, Optuna, and the `aac-metrics` package for evaluation. We use MusicCaps [6] with a fixed three-way random split: a held-out test set (n=529, final metrics only), a validation set (n≈495) for checkpoint selection and hyperparameter search, and the remainder for training (n≈4,455). The split seed is fixed across all conditions, so every model and seed is scored on the identical test set.

A single shared harness scores all conditions with BLEU-1 and BLEU-4 [19], METEOR [20], ROUGE-L [21], CIDEr-D [22], SPICE [23], SPIDEr [24], SBERT-sim, FER and FENSE. FENSE [25] is our primary metric: it combines a Sentence-BERT [26] semantic similarity with a fluency-error penalty (FER), making it sensitive to the semantic correctness a listener cares about. No post-processing is applied by default; sentence trimming is treated as an explicit decoding ablation (Section IV-D).

To judge whether a delta is real, we retrain each baseline with three seeds $(42, 43, 44)$, varying only initialisation, shuffling and dropout while holding the split fixed. The standard deviation of each metric across seeds is our *noise floor*; deltas smaller than it are not interpreted as effects.

B. Baseline comparison

Table I reports greedy-decoding test metrics with per-model noise floors. GPT-2 leads on nine of the ten metrics, including the primary FENSE (0.3998 vs. 0.3954); the exception is FER, where T5 is lower (better), though that gap (0.023) sits inside T5's own FER noise floor (±0.0511). The FENSE margin (0.004) is likewise smaller than either model's FENSE noise floor (±0.0137 for GPT-2, ±0.0345 for T5). We therefore do *not* read it as evidence that one architecture is better at music captioning under this recipe. The noise floor is also ∼2.5× larger for T5 than for GPT-2, so the encoder–decoder is noticeably noisier run-to-run and each model must be judged against its *own* variance.

C. Training stages and projection architecture

In Stage 1, where only the projection is trained, the projection configurations spread widely in validation loss (a range of 0.38 for GPT-2 and 0.83 for T5). After Stage 2 that spread collapses to 0.03 for both models — an 11× reduction for GPT-2 and 31× for T5: full fine-tuning largely erases the architectural differences introduced by the projection. The Stage-2 curves also show early overfitting, with validation loss bottoming within a few epochs, which early stopping handles.

Consistently with that collapse, varying prefix length ($\{4, 16\}$ around 8), dropout ($\{0.1, 0.5\}$ around 0.3) and projection depth ($\{1, 3\}$ around 2) — retraining each from scratch — leaves almost every configuration inside the baseline ± noise-floor band: no projection hyperparameter reliably moves the primary metric for GPT-2. For T5 the prefix variants produce larger FENSE swings, but these coincide with collapses in the fluency-error term rather than genuine semantic gains, as the next section explains.

D. Decoding ablations

Decoding strategies are evaluated on the fixed baseline checkpoint without retraining, isolating generation behaviour. The dominant lever is trimming trailing incomplete sentences: it lifts FENSE from 0.3998 to 0.5796 for GPT-2 and from 0.3954 to 0.5762 for T5.

This gain is, however, *largely an artifact of the metric*. Figure 2 plots FENSE against FER across all decoding

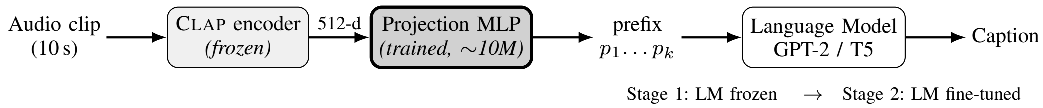

Fig. 1. The CLAPCAP pipeline. A frozen CLAP encoder turns the clip into a single 512-d embedding; a small projection MLP — the only newly added module — maps it to a k-token prefix in the LM embedding space; the LM generates the caption. Stage 1 trains the projection with the LM frozen; Stage 2 additionally fine-tunes the LM.

TABLE I
BASELINE (GREEDY) TEST-SET METRICS PER MODEL, WITH ± RUN-TO-RUN STANDARD DEVIATION OVER THREE SEEDS (THE NOISE FLOOR). BEST PER COLUMN IN BOLD; ALL METRICS ARE HIGHER-IS-BETTER EXCEPT FER.

Model	BLEU-1 ↑	BLEU-4 ↑	METEOR ↑	ROUGE-L ↑	CIDEr-D ↑	SPICE ↑	SPIDEr ↑	SBERT-sim ↑	FER ↓	FENSE ↑
GPT-2	**0.3091**	**0.0631**	**0.1265**	**0.2329**	**0.1109**	**0.1091**	**0.1100**	**0.5820**	0.3478	**0.3998**
	±0.0019	±0.0039	±0.0041	±0.0017	±0.0076	±0.0032	±0.0024	±0.0089	±0.0272	±0.0137
T5	0.3003	0.0574	0.1162	0.2275	0.0847	0.0866	0.0857	0.5564	**0.3251**	0.3954
	±0.0122	±0.0048	±0.0036	±0.0027	±0.0181	±0.0124	±0.0149	±0.0170	±0.0511	±0.0345

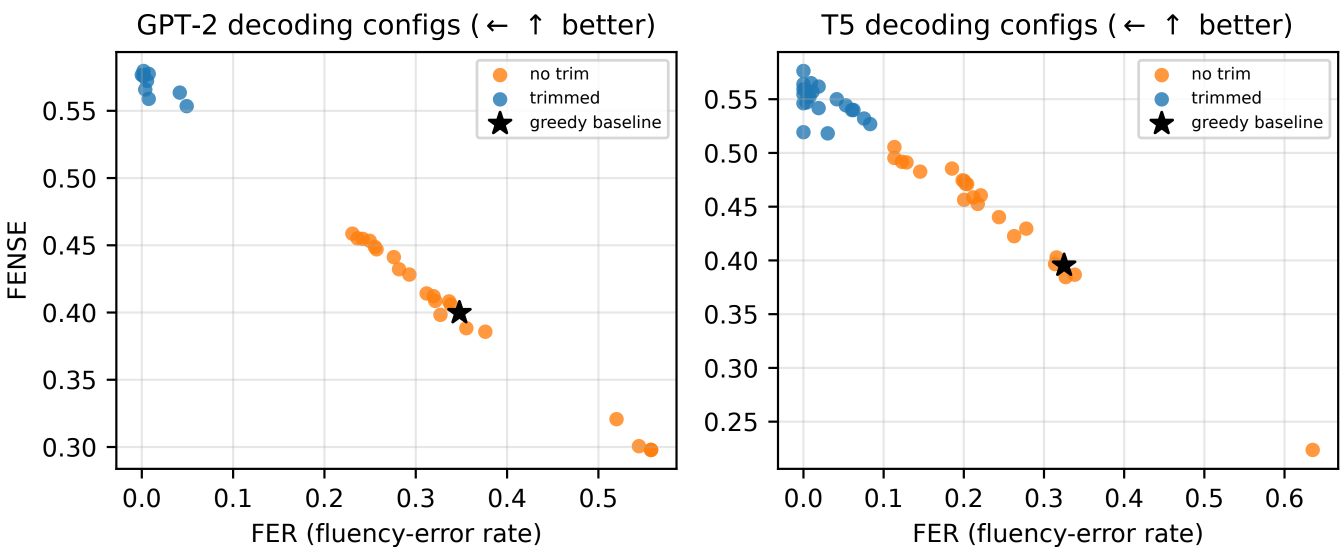

Fig. 2. FENSE vs. FER over all decoding configurations. The near-perfect inverse trend shows that FENSE here is dominated by the fluency-error term: trimmed configurations (low FER) score high FENSE without improving semantic similarity. The star is greedy decoding.

configurations: the relationship is almost perfectly inverse, and the trimmed configurations simply drive FER to near zero while the lexical metrics barely move. Trimming removes the fluency penalty inside FENSE without making captions semantically more correct. We report it as a useful decoding choice for producing clean captions, but caution against reading the headline FENSE jump as a semantic improvement.

E. Audio-encoder ablation

The second axis keeps the language model fixed (GPT-2) and swaps the frozen CLAP encoder for music-specialised alternatives, re-extracting the cached embeddings and re-running the learning-rate search per encoder while leaving the projection, training protocol and evaluation untouched. Because Stage 2 erases upstream differences (Section IV-C), we run this comparison at Stage 1, where the frontend signal is most exposed, and replicate every encoder over the three noise-floor seeds.

Against the general-audio baseline `clap-htsat-unfused` (0.264 ± 0.018 FENSE) we compare two music CLAP checkpoints, two MERT variants [27], MusicFM [28], and the self-supervised MuQ and MuQ-MuLan [29]. Only two encoders clear the baseline band: `clap-music-speech` (0.325 ± 0.027) and `muq-mulan` (0.321 ± 0.008). The remaining five — including the frame-level self-supervised MERT, MusicFM and MuQ, the last of which leads the MARBLE music-understanding benchmark — are not separable from general CLAP at this stage. What survives is therefore the *pooled joint audio–text* pretraining shared by the two clearers, not raw self-supervised acoustic features: at the frozen-projection stage, a CLAP-style clip embedding is what the projection can exploit.

F. Vocalist bias

The models reliably recover genre, instrumentation and mood, but two failure modes recur: on non-music audio the pipeline still emits a confident music caption, and greedy decoding occasionally degenerates into repetition. Most strikingly, both models tend to insert "a male vocalist singing melodically" even when the reference states there is no singer.

The bias is *not* a blind spot. GPT-2 mentions vocals on 95% of clips whose reference describes singing, but also on 43% of clips marked instrumental (T5: 82% / 29%) — both track vocal presence well above chance while skewing toward false positives. The skew mirrors the training captions, of which 53% mention vocals and in which male vocalists outnumber female ones 2.2:1; the phrase "male vocalist" appears in 10% of training captions but in roughly half of the generated ones. Gender is worse: on references describing a *female* singer, T5 outputs "male" 89% of the time and GPT-2 44%, while male references are almost never misgendered (2–4%). These rates persist across decoding configurations. Probing the frozen CLAP embeddings with zero-shot text prompts separates vocal from instrumental at 74% accuracy and vocalist gender at 76%, so the embedding carries the cue but noisily; since both LMs read the *same* embedding, the gap between GPT-2 (56%) and T5 (11%) in female-vocalist accuracy shows the failure is largely model-side. The bias is caption-prior amplification under uncertain audio evidence, not a missing audio cue.

G. Comparison with reference captioners

To place the pipeline against existing systems without an invalid cross-dataset comparison, we score zero-shot reference captioners on the *same* MusicCaps test split through the same harness. These models ingest raw audio and bypass the CLAP–projection–LM design, so they are an external reference, not controlled ablation rows. They span general audio-LMs — Audio Flamingo (the AF-Next captioner checkpoint) [11], Qwen2-Audio [10] and a Qwen3-Omni captioner [30] — and music-domain captioners — LP-MusicCaps [8] and BLAP [9]. Figure 3 summarises the primary metric.

Leakage is the central caveat. Several references may have seen MusicCaps in training: Qwen2-Audio attains the highest FENSE (0.665) but *sweeps every metric including the lexical ones*, the signature of training on the target captions; the MusicCaps-fine-tuned LP-MusicCaps (transfer) reaches FENSE 0.702 with a CIDEr-D of 3.733 — an order of magnitude above any honest row — which is train-on-test memorisation, since $\sim$48% of our random test split falls in its official-train set. We mark such rows with a dagger and, where a model was fine-tuned on MusicCaps-train, additionally report a leakage-free *held-out* score on the MusicCaps-official-eval subset (n=284).

Among *leakage-free* rows the lightweight pipeline is competitive: GPT-2 at its best decoding configuration reaches FENSE 0.580, the highest non-leaked score, ahead of every clean reference — LP-MusicCaps (pretrain, MSD-only) at 0.378, BLAP held-out at 0.476, and LP-MusicCaps transfer held-out at 0.543. The general audio-LMs are fluent but off-style: Audio Flamingo scores well on semantics (FENSE 0.440, SBERT-sim 0.586) yet loses every lexical metric to the trained pipeline, and the 30B Qwen3-Omni captioner is the weakest reference (FENSE 0.396). A small model trained on-target matches the MusicCaps caption style that large zero-shot models miss, and only models that have arguably seen the data score clearly higher.

H. Human and LLM-judge evaluation

The automatic metrics rank GPT-2 and T5 as a statistical tie (Section IV-B). To test whether that tie holds under human and model judgement, we ran a pairwise A-vs-B study. Sixteen clip pairs (six GPT-2-best vs. the human MusicCaps reference, six GPT-2-best vs. T5-best, four reference-vs-mismatched-reference sanity checks) were each judged on two questions: Q1 "which caption is more accurate" and Q2 "which is less wrong", with A/B/tie/can't-tell responses. Pairwise comparison is used rather than absolute Likert scoring because it gives more reliable inter-rater agreement on caption quality.

Ten lay listeners rated every pair; three were removed by a sanity check (agreement with the obvious answer below 0.75), leaving seven. The same pairs were scored by two language-model panels under an identical rubric, both A/B orders averaged to cancel position bias, at temperature zero with three repeats and logged model snapshots: an *audio-grounded* panel that hears the clip (Gemini 2.5 Pro, GPT-audio, Qwen3.5-Omni) and a larger *text-reference-only* panel that sees only the reference caption.

Lay raters are individually noisy but their *consensus* tracks the audio-LLM panel. Inter-human agreement is low (Fleiss κ 0.131 on Q1, 0.050 on Q2), as expected for a non-expert panel; the audio-LLM panel is more consistent (0.392/0.486). Crucially, human-consensus versus audio-LLM-consensus agreement is substantial (Cohen κ 0.643 on Q1, 0.696 on Q2; raw agreement 80%/86%). Win-rates point the same way in every pool (Figure 4): the human reference beats GPT-2 (69% of human and 78% of LLM votes prefer the reference on Q1) and GPT-2 beats T5 (72% human / 71% LLM). So the pipeline does not beat the human reference, and the FENSE tie between GPT-2 and T5 does *not* survive a preference test.

There is no human-versus-machine split in the per-judge agreement matrix: human consensus agrees with the GPT-audio and Qwen judges (Cohen κ up to 0.69/0.76) more than the audio judges agree among themselves, with Gemini 2.5 Pro the dissenting outlier. The dispersion is *within* the model panel, not between humans and models. The text-only panel is small (six pairs, directional only) and one judge degenerates to always answering "tie"; we report these as limitations rather than findings.

V. CONCLUSION

We presented CLAPCAP, a controlled study of lightweight language models for *music* captioning via an audio adaptation of the ClipCap prefix recipe. Holding the frozen CLAP encoder, the projection, the data split and the evaluation fixed, a decoder-only and an encoder–decoder LM are statistically indistin-

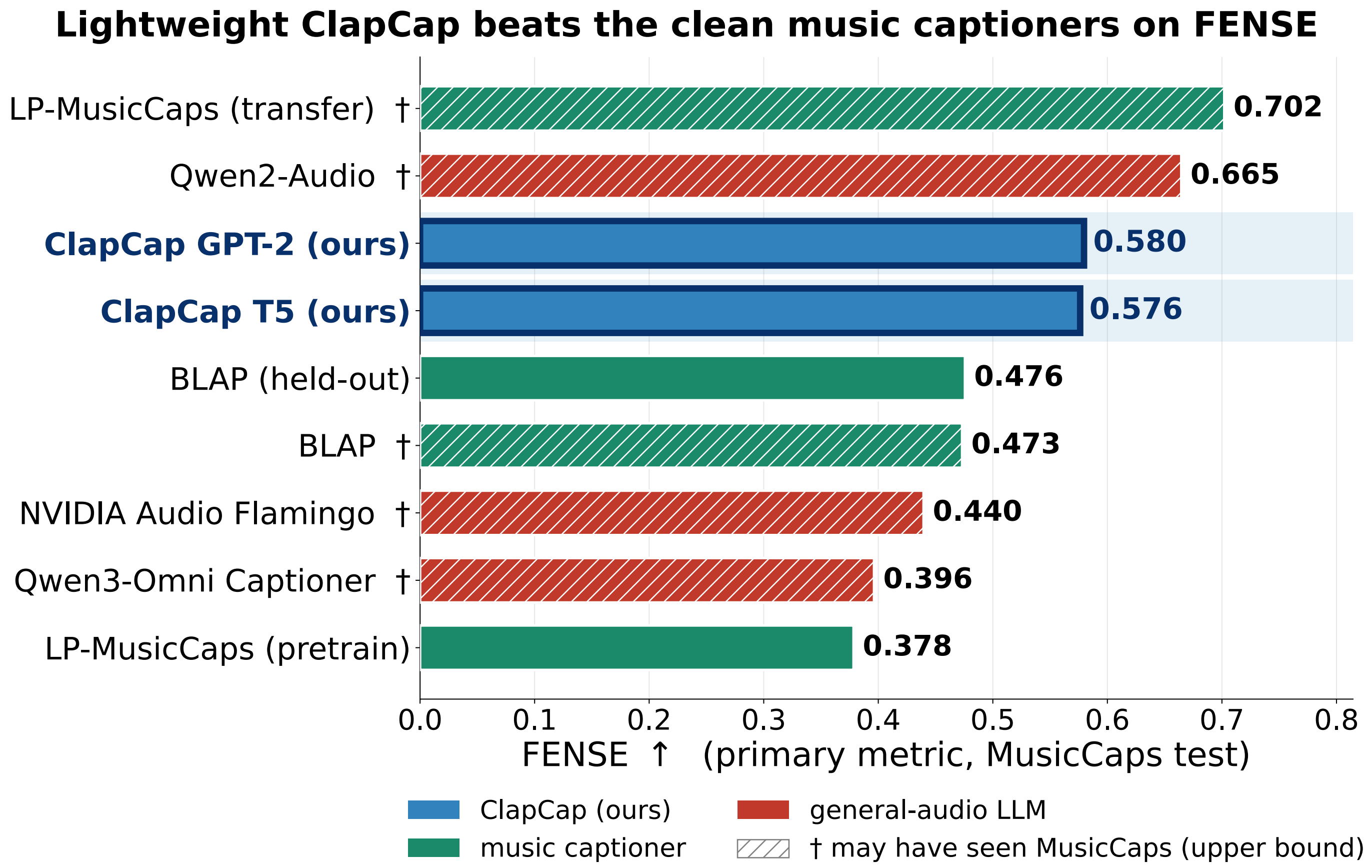

Fig. 3. Primary metric (FENSE) on the shared MusicCaps test split: our lightweight pipeline (GPT-2/T5, highlighted) against the zero-shot reference captioners. Hatched, daggered bars may have seen MusicCaps in training (leakage upper bounds); for the two leaky music captioners the clean held-out number is plotted instead. The pipeline's best decoding configuration is the strongest leakage-free system.

guishable at baseline once run-to-run variance is measured; second-stage fine-tuning erases projection-level differences; and the largest decoding lever improves the primary metric mostly by removing a fluency penalty rather than by improving semantics. A reproducible vocalist bias turns out to be caption-prior amplification under uncertain audio evidence. Swapping the general frontend for music-specialised encoders helps only where the pretraining is pooled and joint audio–text. Against zero-shot reference captioners on the same split, the pipeline at its best decoding setting is the strongest leakage-free system, and a pairwise human and LLM-judge study corroborates the automatic metrics while resolving the GPT-2/T5 tie in GPT-2's favour.

Limitations. The baseline encoder is trained on general audio, and whether a music encoder helps more once the LM is also fine-tuned is left open. Several reference captioners may have been trained on MusicCaps; we dagger them and report held-out scores where possible, but their upper-bound numbers should not be read as clean. The human study uses ten lay listeners, not music experts, with low individual agreement, so the agreement results are directional. Scaling the LM is not free in this low-data regime ($\sim$4.5k clips): a billion-parameter decoder fine-tuned end-to-end overfits within a single epoch — the same overfitting ClipCap [3] reports, amplified by scale. Finally, MusicCaps provides a single reference per clip, which mechanically depresses n-gram metrics.

Future work. Extending the controlled comparison to billion-parameter decoders with a gentler fine-tuning schedule; carrying the encoder ablation into Stage 2 and to a LoRA-fine-tuned domain captioner; and audio-grounded metrics that score a caption against the signal rather than a single text reference.

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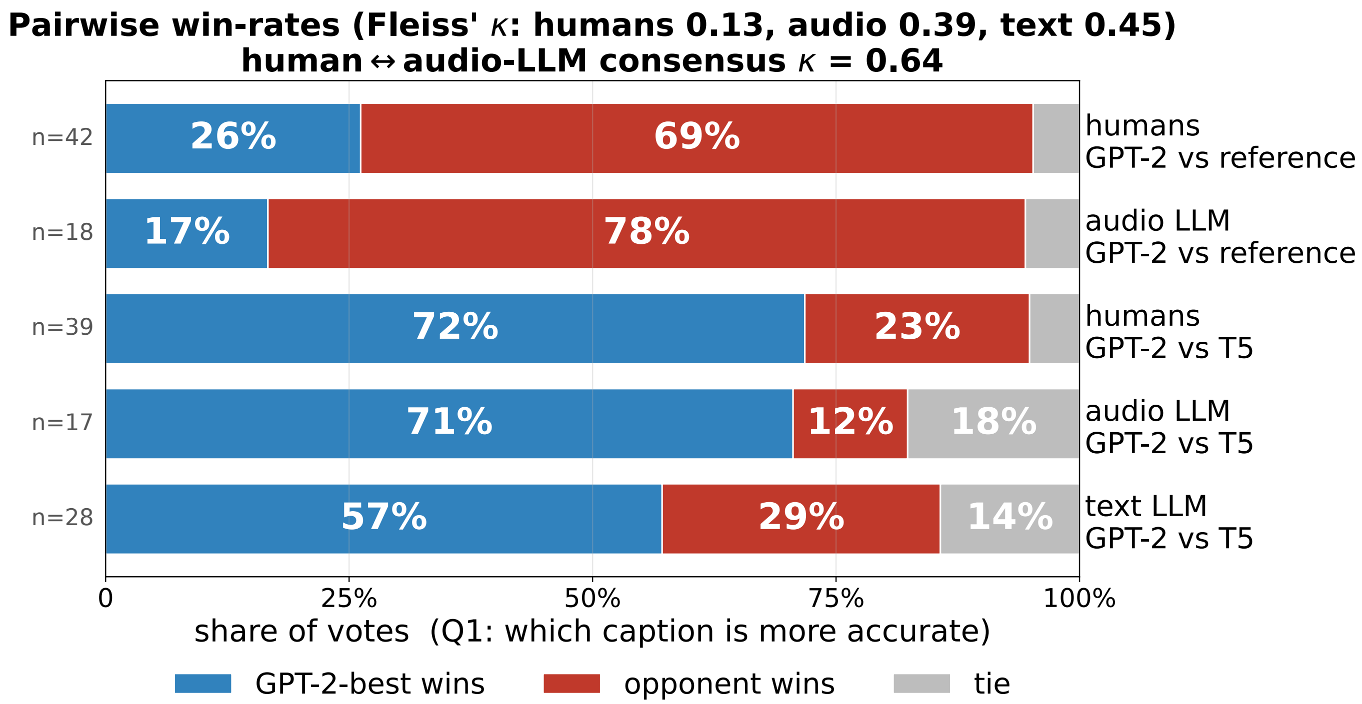

Fig. 4. Pairwise A/B win-rates by rater pool. The reference-vs-GPT-2 block (the reference wins) and the GPT-2-vs-T5 block (GPT-2 wins) point the same direction for the human, audio-LLM and text-LLM pools. Bar counts n are the decisive (non-tie) votes.

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